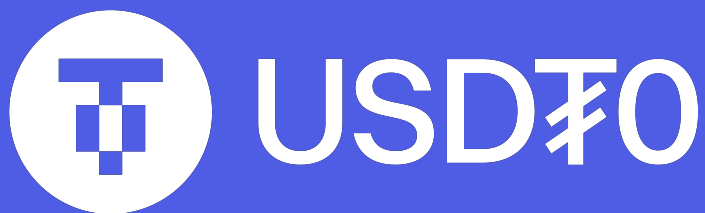


USDT0 Tempo Network Deployment Review



February 20, 2026

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Summary

Type	Stablecoins	Total Issues	0 (0 resolved)
Timeline	From 2026-02-18 To 2026-02-19	Critical Severity Issues	0 (0 resolved)
Languages	TypeScript	High Severity Issues	0 (0 resolved)
		Medium Severity Issues	0 (0 resolved)
		Low Severity Issues	0 (0 resolved)
		Notes & Additional Information	0 (0 resolved)

Scope

OpenZeppelin performed a bytecode verification of the USDT0 Omnichain Fungible Token (OFT) deployment on the Tempo Network. The verification was performed using the implementation addresses provided within the [Everdawn-Labs/usdt0-oft-contracts](#) repository in [pull request #183](#) at commit [1d706ad](#).

This verification was performed to ensure that the deployed bytecode of the USDT0 OFT implementation on Tempo and the related contracts matched the expected in-repository version.

System Overview

USDT0 is a cross-chain stablecoin that uses LayerZero's Omnichain Fungible Token (OFT) design to facilitate cross-chain transfers. The `0UpgradeableTempo` contract extends `OFTCoreUpgradeable` and serves as the bridge adapter for USDT0 on the Tempo Network.

The deployment on the Tempo Network differs from previous deployments in two notable ways:

- Tempo implements ERC-20 tokens as precompiles following the [TIP-20 specification](#). The TIP-20 token standard incorporates role-based access control and integrates with the [TIP-403 specification](#), which enables granular transfer policy management, including KYC/KYB operations.
- The Tempo Network uses an ERC-20 token for gas payments rather than a native currency. As a result, the `0UpgradeableTempo` contract overrides the `_payNative` function to transfer the native fee token (LayerZeroEndpointDollar) via `safeTransferFrom` to the LayerZero endpoint, instead of forwarding native `msg.value`.

Deployment Verification

USDT0

Contract	Address
OFT Proxy	<code>0xaf37E8B6C9ED7f6318979f56Fc287d76c30847ff</code>
OFT Implementation	<code>0x01bFF41798a0BcF287b996046Ca68b395DbC1071</code>
DefaultProxyAdmin	<code>0xe7cd86e13AC4309349F30B3435a9d337750fC82D</code>
TIP-20 Token (USDT0)	<code>0x20C0000000000000000000000000000014f22CA97301EB73</code>

Trust Assumptions and Privileged Roles

The system relies on several privileged roles across two layers: the OFT contract and the underlying TIP-20 token.

OFT Contract Roles

The `0UpgradeableTempo` contract inherits an `owner` privileged role that controls LayerZero OFT configuration, including peer chain mappings and messaging library settings. The Everdawn Safe multisig (`0x4DFF9b5b0143E642a3F63a5bcf2d1C328e600bf8`) currently holds this role. This same account is the owner of the `DefaultProxyAdmin` contract, which controls implementation upgrades for the OFT proxy.

It should be noted that the USDT0 token has been whitelisted by the `LayerZeroEndpointDollar` token to allow token transfers.

TIP-20 Token Roles

The underlying TIP-20 USDT0 token precompile defines the following roles:

- `ISSUER_ROLE`: Authorized to mint new tokens. The OFT proxy requires this role to credit incoming cross-chain transfers.
- `PAUSE_ROLE`: Authorized to pause token transfers.
- `UNPAUSE_ROLE`: Authorized to unpause token transfers.
- `BURN_BLOCKED_ROLE`: Controls burn restrictions on the token.

In addition, the TIP-403 specification allows a `DEFAULT_ADMIN_ROLE` to configure transfer policies on the token, enabling granular control over which accounts may send and receive tokens.

Conclusion

OpenZeppelin performed a bytecode verification and an OFT configuration review of the USDT0 Tempo network deployments as contained within [pull request #183](#).

The deployment under review has notable differences from other deployments. This is because the USDT0 token has been deployed as a TIP-20 contract which uses the [TIP-403 specification](#) for role-based access control. The LayerZero Endpoint configuration was also found to be non-default which necessitated implementation changes in the [0UpgradeableTempo](#) contract. Overall, the deployment review did not identify any issues.

The Everdawn team is appreciated for their responsiveness throughout the review process.

Appendix

Issue Classification

OpenZeppelin classifies smart contract vulnerabilities on a 5-level scale:

- Critical
- High
- Medium
- Low
- Note/Information

Critical Severity

This classification is applied when the issue's impact is catastrophic, threatening extensive damage to the client's reputation and/or causing severe financial loss to the client or users. The likelihood of exploitation can be high, warranting a swift response. Critical issues typically involve significant risks such as the permanent loss or locking of a large volume of users' sensitive assets or the failure of core system functionalities without viable mitigations. These issues demand immediate attention due to their potential to compromise system integrity or user trust significantly.

High Severity

These issues are characterized by the potential to substantially impact the client's reputation and/or result in considerable financial losses. The likelihood of exploitation is significant, warranting a swift response. Such issues might include temporary loss or locking of a significant number of users' sensitive assets or disruptions to critical system functionalities, albeit with potential, yet limited, mitigations available. The emphasis is on the significant but not always catastrophic effects on system operation or asset security, necessitating prompt and effective remediation.

Medium Severity

Issues classified as being of medium severity can lead to a noticeable negative impact on the client's reputation and/or moderate financial losses. Such issues, if left unattended, have a moderate likelihood of being exploited or may cause unwanted side effects in the system.

These issues are typically confined to a smaller subset of users' sensitive assets or might involve deviations from the specified system design that, while not directly financial in nature, compromise system integrity or user experience. The focus here is on issues that pose a real but contained risk, warranting timely attention to prevent escalation.

Low Severity

Low-severity issues are those that have a low impact on the client's operations and/or reputation. These issues may represent minor risks or inefficiencies to the client's specific business model. They are identified as areas for improvement that, while not urgent, could enhance the security and quality of the codebase if addressed.

Notes & Additional Information Severity

This category is reserved for issues that, despite having a minimal impact, are still important to resolve. Addressing these issues contributes to the overall security posture and code quality improvement but does not require immediate action. It reflects a commitment to maintaining high standards and continuous improvement, even in areas that do not pose immediate risks.